# Numerical Methods

Questions 1–49

**2.27 — Question 1.** Show that the following equations have a solution in the given interval:

a) $x^3 - 4x - 1 = 0$ for $x \in [2, 3]$
b) $2 + x - e^x = 0$ for $x \in [1, 2]$
c) $x - 4 - 2\ln x = 0$ for $x \in [8, 9]$
d) $1 - x \times 5^x = 0$ for $x \in [0, 1]$

**2.27 — Question 2.**

$$\mathrm{f}(x) = \frac{1}{x - 3}$$

a) Find f(2) and f(4)
b) Sketch f($x$)
c) Explain why a change in sign of f($x$) between $x = 2$ and $x = 4$ is not sufficient to show that there is a root.

**2.27 — Question 3.** Draw a sketch for each of these situations for a continuous function:

a) no change of sign, no real roots.
b) no change of sign, one real root.
c) no change of sign, two real roots.

**2.27 — Question 4.** Draw a sketch for each of these situations for a discontinuous function:

a) change of sign, no real roots.
b) change of sign, one real root
c) change of sign, two real roots
d) no change of sign, no real roots
e) no change of sign, one real root
f) no change of sign, two real roots

**2.27 — Question 5. Prerequisite knowledge for Q5: Radians**

$$f(x) = \frac{1}{2x} - \tan x$$

a) Find f(−1), f(1) and f(2)
b) Explain if there are any roots between $x = -1$ and $x = 1$, and between $x = 1$ and $x = 2$

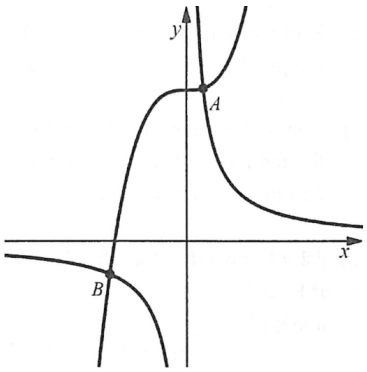
**2.27 — Question 6.**

a) Show that the equation $e^x - \frac{1}{x} = 0$ has a root between $x = 0.5$ and $x = 1$
b) Explain whether or not there is a root between $x = -1$ and $x = 1$

**2.27 — Question 7.** Sketch the graphs of $y = e^x$ and $y = 4 - x^2$ showing that $e^x = 4 - x^2$ has two solutions.

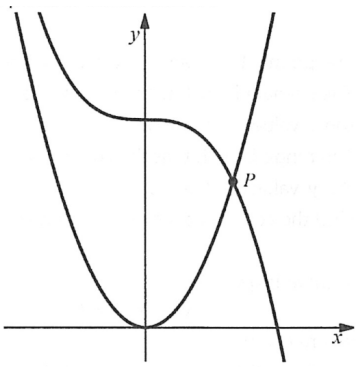
**2.27 — Question 8.** Sketch the graphs of $y = \sqrt{x-1}$ and $y = \frac{1}{2}x$ showing that $\sqrt{x-1} = \frac{1}{2}x$ has one solution.

**2.27 — Question 9.** The diagram shows the graphs of $y = \frac{1}{x}$ and $y = x^3 + 3$




a) Show that the $x$-coordinates of point $A$ and point $B$ satisfy $x^4 + 3x - 1 = 0$
b) Show that the $x$-coordinate of point $A$ lies between $x = 0$ and $x = 1$
c) Show that the $x$-coordinate of point $B$ lies between $x = -2$ and $x = -1$

**2.27 — Question 10.** The diagram shows the graphs of $y = x^2$ and $y = -\frac{1}{4}x^3 + 4$ with point of intersection $P$.




a) Show that the $x$-coordinate of $P$ is the solution of $x^3 + 4x^2 - 16 = 0$
b) Show that the $x$-coordinate of point $P$ lies between 1.67855 and 1.67865
c) Given that the $x$-coordinate of $P$ is $\alpha$, write down the value of $\alpha$ to 4 decimal places.

**2.28 — Question 1.** Show that the equation $x^2 + x - 4 = 0$ may be arranged in to each of the following:

a) $x = \sqrt{4 - x}$

b) $x = \frac{4}{x+1}$

**2.28 — Question 2.** Show that the equation $x^3 + 2x - 5 = 0$ may be arranged in to each of the following:

a) $x = \sqrt[3]{5 - 2x}$

b) $x = \frac{5}{x^2+2}$

**2.28 — Question 3.** Given that $f(x) = x^3 - x^2 - 2x - 3$

a) Show that the equation $f(x) = 0$ may be written in the form $x = \sqrt{\frac{3+2x}{x-1}}$

b) Use your answer to part a) to write a recurrence relation for finding a solution to $f(x) = 0$ then starting with $x_1 = 2$, find a solution to 4 decimal places.

**2.28 — Question 4.** Given that $f(x) = x^2 \ln x - 5x$

a) Show that $f(x) = 0$ has a solution between $x = 3$ and $x = 4$

b) Show that the equation $f(x) = 0$ may be written in the form $x = \sqrt{\frac{5x}{\ln x}}$

c) Use your answer to part b) to write a recurrence relation for finding a solution to $f(x) = 0$ then starting with $x_1 = 3$, find a solution to 4 decimal places.

**2.28 — Question 5. Prerequisite knowledge for Q5 b): Radians** For each of the following recurrence relations, find $x_2$, $x_3$ and $x_4$ giving your answer to 4 decimal places:

a) $x_{n+1} = \sqrt{x_n + 3}$ where $x_1 = 5$

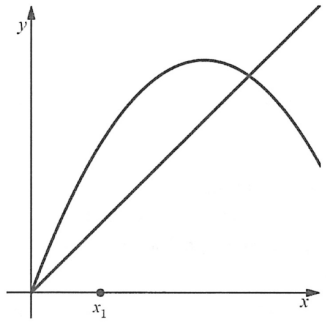
b) $x_{n+1} = 2\sin(x_n)$ where $x_1 = 1$ ($x_n$ is measured in radians)

c) $x_{n+1} = 8 - \frac{1}{10}e^{0.4x_n}$ where $x_1 = 0$

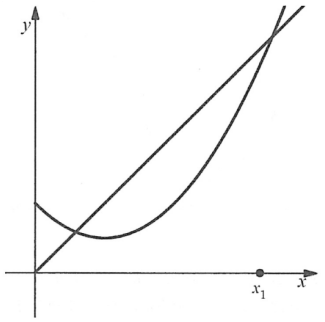
d) $x_{n+1} = 0.1x_n^3 - 0.5x_n + 2$ where $x_1 = 2$

**2.28 — Question 6.** For each graph below, draw a diagram to show the convergence or divergence cobweb or staircase starting at the indicated point, labelling each of $x_2$, $x_3$ and $x_4$.
Identify whether it is a convergent or a divergent cobweb or staircase diagram.

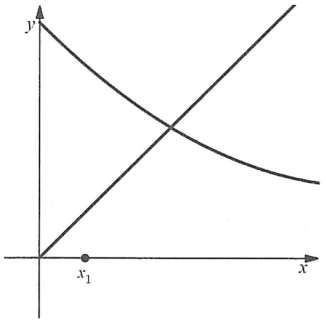
**a)**



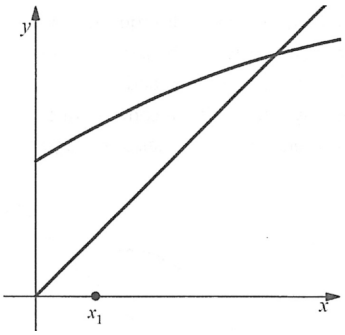

**b)**



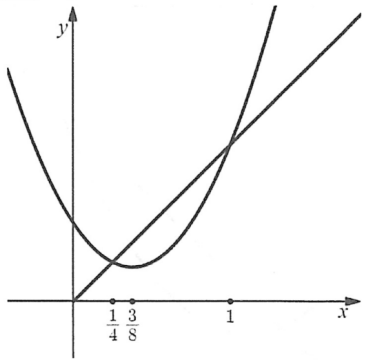

**c)**




**d)**



**2.28 — Question 7.** The diagram shows $y = 2x^2 - 1.5x + 0.5$ and $y = x$ on the same graph. They intersect when $x = \frac{1}{4}$ and $x = 1$.




Describe what happens in terms of a cobweb or staircase diagram for each of the following starting points:

a) $-\frac{1}{4} < x_1 < \frac{1}{4}$
b) $x_1 = \frac{1}{4}$
c) $\frac{1}{4} < x_1 < 1$
d) $x_1 = 1$
e) $x_1 > 1$

**2.28 — Question 8. Prerequisite knowledge for Q8 b): Radians** Each of the following iterative formulae has a sequence which converges to a limit $x = \alpha$. In each case, deduce an interval of width 0.001 in which $\alpha$ lies:

a) $x_{n+1} = 2 + \ln x_n$ where $x_1 = 3$
b) $x_{n+1} = \cos^2(x_n) + 1$ where $x_1 = 1.15$ ($x_n$ is measured in radians)
c) $x_{n+1} = \frac{3}{2}x_n - \frac{1}{10}x_n^4$ where $x_1 = 1$

**2.28 — Question 9.** An iterative formula for finding root $\alpha$ is given by $x_{n+1} = \mathrm{g}(x_n)$. By considering $\mathrm{g}'(\alpha)$ in each of the following cases, explain whether the iterative formula will converge or diverge to the given root (given to 3 decimal places) from a point considered close to the root.

a) $x_{n+1} = x_n^5 - 2x_n^4 - 5$ where $\alpha = 2.273$
b) $x_{n+1} = e^{x_n} - 2$ where $\alpha = 1.146$
c) $x_{n+1} = e^{x_n} - 2$ where $\alpha = -1.841$

**2.28 — Question 10.** Harry is trying to find the values of the three roots of the equation $y = x^3 - 3x^2 - x + 4$. He rearranges the equation into four of the possible iterative formula $x_{n+1} = \mathrm{g}(x_n)$ to find a solution $\alpha$, including finding an initial point:

$$\begin{aligned}
&\text{A}: \quad x_{n+1} = x_n^3 - 3x_n^2 + 4 && x_1 = 2 \\
&\text{B}: \quad x_{n+1} = \sqrt[3]{3x_n^2 + x_n - 4} && x_1 = 2.8 \\
&\text{C}: \quad x_{n+1} = \frac{x_n - 4}{x_n^2 - 3x_n} && x_1 = 1.2 \\
&\text{D}: \quad x_{n+1} = -\sqrt{\frac{x_n^3 - x_n + 4}{3}} && x_1 = -1
\end{aligned}$$

a) Find the three roots using the iterative formulae B, C and D at the given initial points above, giving your answers to 2 decimal places.
b) By considering $\mathrm{g}'(\alpha)$ on iterative formula A, explain why this iterative process does not converge to a root for each of the roots that you found in part a).

**2.28 — Question 11.**

a) Sketch the graphs $y = \frac{1}{x}$ and $y = \sin x$ to show that the equation $\frac{1}{x} = \sin x$ has exactly one solution for $0 < x < \frac{\pi}{2}$

b) Show that the solution to the equation lies between $x = 1$ and $x = 1.2$

c) By using a suitable iterative formula, obtain an approximation solution to $\frac{1}{x} = \sin x$ to 4 decimal places.

**2.28 — Question 12.**

a) Sketch the graphs $y = x$, $y = e^{x-2}$ and $y = 2 + \ln x$ on the same axes, for $0 < x < 4$

b) Explain why the solutions to the equation $x = e^{x-2}$ are also solutions to the equation $e^{x-2} = 2 + \ln x$

**2.28 — Question 13.** The number of tickets sold $N$ for an event in the first 48 hours of sales is modelled by $N = 3t - e^{0.1t}$ where $t$ is the time in hours passed. When the number of tickets sold drops to zero and $t = T$

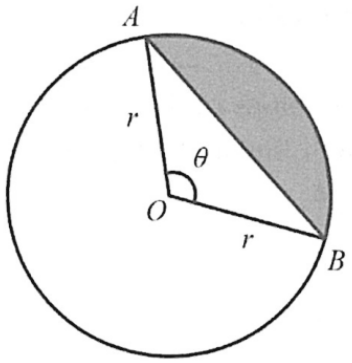
a) Show that $T$ satisfies $T = 10\ln(3T)$

b) Write a recurrence relation to find the solutions to $T = 10\ln(3T)$ and starting with $T_1 = 48$, find a solution to the equation to 3 decimal places.

c) Hence state the number of hours that have passed, to the nearest hour, when the number of tickets being sold is zero.

**2.28 — Question 14. Prerequisite knowledge for Q14: Radians**

A circle with centre O has a chord such that the angle $AOB = \theta$ radians where $0 < \theta < \pi$




The area of the segment cut off by $AB$ is $\frac{1}{6}$ of the area of the circle.

a) Show that $\theta - \sin\theta = \frac{\pi}{3}$

The solution lies between $\theta = 1$ and $\theta = 3$

b) Using an iterative method, find the solution correct to 2 decimal places.

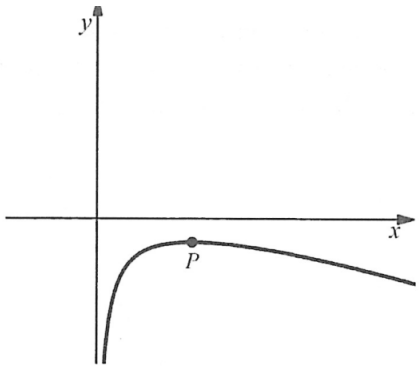
**2.28 — Question 15.** Giving your answers to 3 decimal places, use an iterative method to find the $x$-coordinates of the points of intersection of:

a) $y = x$ and $y = \cos x$
b) $y = x^2$ and $y = x \log_3(0.5x + 1)$

**2.28 — Question 16. Prerequisite knowledge for Q16: Inverse Trigonometric Functions**

a) Sketch the graphs of $y = 3\arctan(2x)$ and $y = x^2$ on the same axes for $-4 \leq x \leq 4$
b) Show that there is a solution to $x^2 - 3\arctan(2x) = 0$ between $x = 1.8$ and $x = 2.2$
c) By using a suitable iterative formula, obtain an approximation solution to $x^2 - 3\arctan(2x) = 0$ to 3 decimal places.

**2.28 — Question 17. Prerequisite knowledge for Q17: Differentiation of** $\ln x$ The diagram shows the graph of $y = -\frac{2x^2+24}{3\sqrt{x}} - 4\ln x$. The stationary point of the curve is labelled $P$.




a) Show that the point $P$ satisfies $0 = \sqrt{x} - \frac{1}{4}x^2\sqrt{x} - x$
b) Use the iteration formula $x_{n+1} = \sqrt{x_n} - \frac{1}{4}x_n^2\sqrt{x_n}$ starting at $x_1 = 1$ to find the $x$-coordinate of point $P$ to 2 decimal places.

**2.28 — Question 18. Prerequisite knowledge for Q18: Double Angle Formulae** An approximate solution to an equation can be found using the iterative formula $x_{n+1} = \frac{1}{4}\arccos x_n$.

a) By using $x_1 = 0.3$, find $x_2$, $x_3$ and $x_4$ giving your answers to 4 decimal places.
b) Show that the iterative formula may be used to find a solution to $x - 1 + 8\sin^2 x \cos^2 x = 0$

**2.29 — Question 1.** Use the Newton Raphson Method with the value of $x_1$ given to find the next approximation, $x_2$, to find a root of each of the following equations, giving your answers to 2 decimal places:

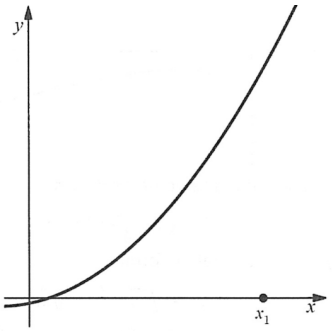
a) $x^4 + x^2 - 80 = 0$ using $x_1 = 3$
b) $x^3 + 2x^2 - 5x - 4 = 0$ using $x_1 = 2$
c) $e^x + x - 4 = 0$ using $x_1 = 1.5$

**2.29 — Question 2.** Use the Newton Raphson Method with the value of $x_1$ given to find the next approximation, $x_2$, to find a solution of each of the following equations, giving your answers to 2 decimal places:

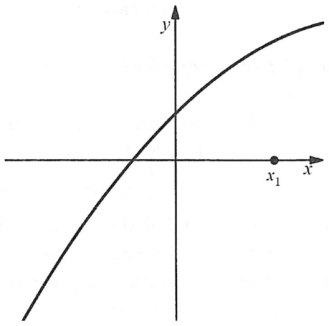
a) $2x^4 + x^3 = 5$ using $x_1 = 1.1$
b) $\frac{1}{6}x^6 - 8x = 4$ using $x_1 = 2$
c) $e^{2x} - x = 3$ using $x_1 = 0.65$

**2.29 — Question 3.** For each graph below, draw a diagram to show the Newton-Raphson starting at the indicated point, labelling each of $x_2$, $x_3$ and $x_4$.

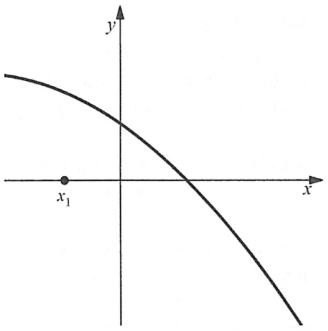
**a)**



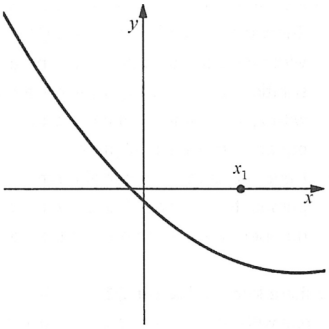

**b)**



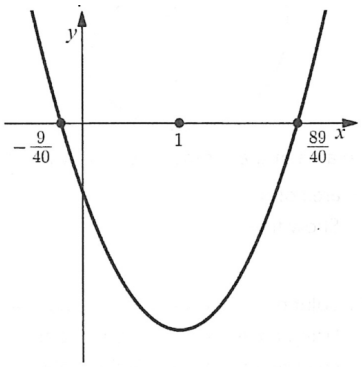

**c)**




**d)**



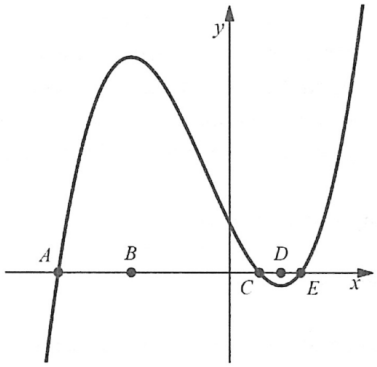

**2.29 — Question 4.** The diagram shows $y = 2x^2 - 4x - 1$ with the $x$-values of the roots and the minimum point labelled on the $x$-axis.




Describe what happens when the Newton-Raphson iterative method is used with each of the following starting points:

a) $x_1 < 1$
b) $x_1 = 1$
c) $x_1 > 1$

**2.29 — Question 5.** The diagram shows $y = x^3 + x^2 - 3x + 1$ with the $x$-values of the roots and the turning points labelled on the $x$-axis.




Describe what root is reached when using the Newton-Raphson method for each of the following starting points:

a) $x_1 = A$
b) $x_1 < B$
c) $x_1 = B$
d) $B < x_1 < D$
e) $x_1 = C$
f) $x_1 = D$
g) $x_1 > D$
h) $x_1 = E$

**2.29 — Question 6.** By considering the root of the equation $x^3 - 20 = 0$ near $x = 3$, use Newton-Raphson method to find $\sqrt[3]{20}$ correct to 3 decimal places.

**2.29 — Question 7.** The equation $3x^3 + 4x^2 - 3 = 0$ has one real root.

a) By considering a change in sign, show that there is a root between $x = 0$ and $x = 1$.

b) Show that the Newton-Raphson formula for finding the root can be written in the form:

$$x_{n+1} = \frac{6x_n^3 + 4x_n^2 + 3}{9x_n^2 + 8x_n}$$

c) Use the Newton-Raphson method with $x_1 = 1$ to find $x_2$ and $x_3$ to 4 decimal places.

d) Explain why the method fails when $x_1 = 0$ or $x_1 = -\frac{8}{9}$.

**2.29 — Question 8. Prerequisite knowledge for Q8: Arithmetic Series** An arithmetic series has first term 2 and common difference 3. The sum of the first $n$ terms is $\frac{12}{n}$.

a) Show that $n$ satisfies the equation $3n^3 + n^2 - 24 = 0$.

b) Apply the Newton-Raphson method to the equation above using $n_1 = 2$ to obtain an answer correct to 4 significant figures.

**2.29 — Question 9. Prerequisite knowledge for Q9: Geometric Series** A geometric series has first term 2 and common ratio $x$. The sum of the first 8 terms is 18.

a) Show that $x$ satisfies the equation $x^8 - 9x + 8 = 0$.

b) Apply the Newton-Raphson method to the equation above using $x_1 = 0.9$ to obtain an answer correct to 4 significant figures.

c) Hence determine the sum to infinity of the geometric series giving your answer to 3 significant figures.

**2.29 — Question 10. Prerequisite knowledge for Q10: Modulus Functions**

a) Sketch the graph of $y = 5 - \frac{1}{x^2}$ and $y = 4 - |2x|$ on the same axes showing clearly the points of intersection.

b) Show that $\alpha$, the negative root of $5 - \frac{1}{x^2} = 4 - |2x|$ satisfies the equation $2x^3 - x^2 + 1 = 0$.

c) Hence show that $\alpha$ may be found using a Newton-Raphson iterative formula which may be written in the form

$$x_{n+1} = \frac{-4x_n^3 + x_n^2 + 1}{2x_n - 6x_n^2}$$

d) Use the formula starting with $x_1 = -1$ to find an approximation for $\alpha$ to 3 decimal places.

e) State the starting values at which the Newton-Raphson iterative formulae will fail.

**2.29 — Question 11. Prerequisite knowledge for Q11–Q14: Differentiation of Standard Functions**

a) Show that $\sin x = 2 - x$ has a solution between $x = 1$ and $x = 2$.

b) Using the Newton-Raphson method and $x_1 = 0.4$ find the solution to $\sin x = 2 - x$ to 3 significant figures.

**2.29 — Question 12.**

a) Draw a sketch of $y = \ln x$ and $y = 3 - x$ on the same axes for $0 \leq x \leq 3$.

b) By considering a suitable change in sign, show that the intersection of the two graphs lies between $x = 1$ and $x = 3$.

c) Use the Newton-Raphson method with $x_1 = 2$ to find the solution to 3 significant figures.

**2.29 — Question 13.** The equation $x^2 + \ln(5x) = 0$ has one real root.

a) Show that there is a root between $x = 0.1$ and $x = 0.2$.

b) Show that the Newton-Raphson formula can be written in the form:

$$x_{n+1} = \frac{x_n^3 + x_n - x_n \ln(5x_n)}{2x_n^2 + 1}$$

c) Use the Newton-Raphson method starting with $x_1 = 0.1$ to find $x_2$, $x_3$ and $x_4$ to 4 decimal places.

d) Explain why the Newton-Raphson method can never fail to find the root.

**2.29 — Question 14.** A curve has equation $y = 3x - x^2 + 10\ln(kx + 1)$, $x > 0$ where $k$ is a positive constant. It has a local maximum when $x = 3$.

a) Show that $k = 3$.

b) Show that the curve intersects the axes between $x = 7$ and $x = 8$.

c) Use the Newton-Raphson method with initial value $x_1 = 7$ to find the $x$-coordinate of this point of intersection to 3 decimal places.

**2.29 — Question 15. Prerequisite knowledge for Q15–Q20: Product Rule**

a) Show that $x \sin x = 1$ has a solution between $x = 1$ and $x = 2$.

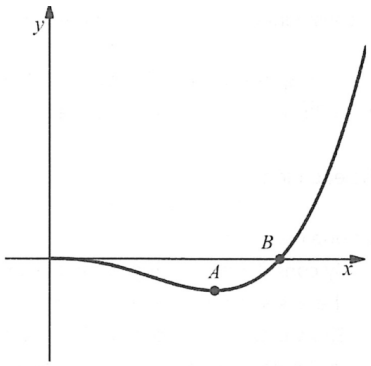
b) Using the Newton-Raphson method and $x_1 = 1$ find a solution to $x \sin x = 1$ giving your answer correct to 3 significant figures.

**2.29 — Question 16.** Amy is trying to solve $2^x x^2 = 3$. She sets $f(x) = 2^x x^2 - 3$ and decides to use the Newton-Raphson method, starting at $x_1 = 0$.

a) Explain why she will not succeed in finding the solution.

b) Use the Newton-Raphson method at $x_1 = 1$ to find $x_3$, an approximation of the root giving your answer to 3 decimal places.

**2.29 — Question 17.** The diagram below shows $y = x^3 \ln x$, $x > 0$ with the local minimum and the root labelled as $A$ and $B$ respectively.



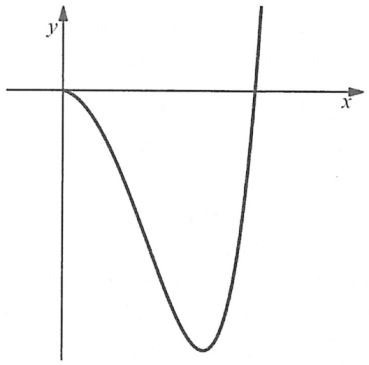

a) State the coordinates of point $B$.
b) Show that $\frac{d^2y}{dx^2} = 5x + 6x \ln x$.
c) By using Newton-Raphson's method and starting at $x_1 = 1$, show that the $x$-coordinate of the minimum point $A$ is 0.717 to 3 significant figures.

**2.29 — Question 18.** The equation $x^2e^x - 3x + 1 = 0$ has two roots.
a) Show that there is a root between $x = 0.4$ and $x = 0.5$.
b) Show that the Newton-Raphson formula to find the roots of $x^2e^x - 3x + 1 = 0$ can be written in the form:

$$x_{n+1} = \frac{x_n^3 e^{x_n} + x_n^2 e^{x_n} - 1}{x_n^2 e^{x_n} + 2x_n e^{x_n} - 3}$$

c) Use the above iterative formula with starting point $x_1 = 0.4$ to find $x_2$ and $x_3$ giving your answers to 3 significant figures.
d) Use the Newton-Raphson method starting at $x_1 = 0.6$ to find the root of $x^2e^x + 2xe^x - 3 = 0$ giving your answer to 3 significant figures.
e) State the relevance of the result you found in part d) to with regards the curve $y = x^2e^x - 3x + 1$.

**2.29 — Question 19.** The curve $y = x \times 3^x - 8x^2 - 2x$, $x > 0$ has a single stationary point which is close to $x = 2.2$.




a) Show that the $x$-coordinate of this stationary point satisfies the equation $3^x + (3^x \ln 3 - 16)x - 2 = 0$.
b) Show that the Newton-Raphson iterative formula for finding the $x$-value of the stationary point is

$$x_{n+1} = x_n - \frac{3^{x_n} + (3^{x_n} \ln 3 - 16)x_n - 2}{(2 \ln 3 + (\ln 3)^2 x_n)3^{x_n} - 16}$$

c) Find the $x$-coordinate of the stationary point giving it correct to 3 significant figures.

**2.29 — Question 20.** The number of views per day of a video $N$ on a social media platform may be modelled by

$$N = 8e^{10-0.25t} \sin(3.15 - 0.19t)$$

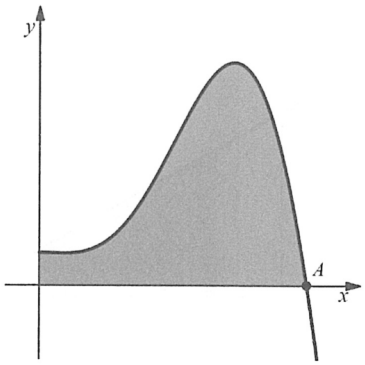
where $t$ is the number of days passed since the video was shared. The maximum of the model is found to occur when $t \approx 3.4646$.

a) Show that that the number of days passed when it reaches the maximum number of views per day is approximately 3.4646.
b) Use Newton-Raphson's method to find the number of days passed to the nearest hour when the views per day is zero.

Comment on the validity of the model

c) when 0 days have passed
d) after the number of days calculated in part b) has passed.

**2.29 — Question 21. Prerequisite knowledge for Q21: Trapezium Rule** The diagram below shows $y = 0.1 + 0.1x^2 \sin(x - 0.5)$, $x > 0$ with the root labelled as $A$.




The shaded region is bounded by $x = 0$, $y = 0$ and where the graph meets the $x$-axis.

a) Show that the $x$-coordinate of the point $A$ lies between $x = 3$ and $x = 4$.

b) Use the Newton-Raphson method, to find an approximation for the $x$-coordinate of $A$ giving your answer to 3 decimal places.

c) Using your $x$-coordinate for $A$, use the trapezium rule with 4 strips to find an approximate value for the shaded area. Give your answer to 1 decimal place.